

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Constraint-Based Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO5: *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.*
(BL5)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5

Total Marks: 50

Code of Conduct

I **SWETA R (192511408)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: SWETA R

Assessment Tasks

Constraint-Based Problem Solving

1. Design an HDFS-based storage solution for a media platform under the constraints that data availability must remain above 99.9%, replication factor cannot exceed 3, and storage growth must stay cost efficient. Justify your design.
2. A MapReduce workflow must process log data within a 20-minute batch window using limited cluster nodes. Propose a job design under these constraints and explain task distribution.
3. A YARN-managed cluster supports ETL, reporting, and machine learning jobs. Design a scheduler policy under the constraints of fairness, priority for ETL, and recovery from node failure.
4. An enterprise needs to ingest data from SAN, NAS, and operational databases into a big data platform while maintaining backup reliability and minimal duplication. Propose a constrained architecture and justify each component.
5. Design an end-to-end Hadoop ecosystem solution under the constraints of secure data movement, high availability, and integration with Apache NiFi or ETL pipelines. Explain how your design satisfies all constraints.

ANSWERS:

1. HDFS-Based Storage Solution for a Media Platform

Problem:

A media platform needs to store large amounts of video, audio, and image data. Data availability must be above **99.9%**, replication must not exceed **3**, and storage costs must remain efficient.

Proposed Solution:

- Use **HDFS** with a **replication factor of 3** for important media data.
- Distribute data blocks across multiple DataNodes to avoid a single point of failure.
- Use **NameNode High Availability** with Active and Standby NameNodes.
- Store frequently accessed data on high-performance storage and older data using **cost-efficient cloud/object storage**.
- Apply data compression and lifecycle policies to reduce storage usage.

Justification:

Replication and NameNode HA provide high availability, while compression, tiered storage, and lifecycle management control storage costs. The replication factor of 3 provides fault tolerance without exceeding the given constraint.

2. MapReduce Workflow for Log Processing

Problem:

A MapReduce workflow must process large log files within a **20-minute batch window** using a limited number of cluster nodes.

Proposed Solution:

- Divide log files into blocks and assign them to multiple **Mapper tasks**.
- Mappers extract required information such as IP address, timestamp, errors, and user activity.
- Use **Combiners** to perform local aggregation and reduce network traffic.
- Reducers aggregate the intermediate results and generate the final report.
- Configure suitable block sizes and run tasks in parallel across available nodes.

Task Distribution:

Input Logs → Mapper Tasks → Combiner → Shuffle & Sort → Reducer Tasks → Output

Justification:

Parallel execution allows multiple log blocks to be processed simultaneously. Combiners reduce data transfer, helping the workflow complete within the 20-minute constraint despite limited resources.

3. YARN Scheduler Policy

Problem:

A YARN cluster must support **ETL, reporting, and machine-learning jobs** while maintaining fairness, giving ETL higher priority, and recovering from node failures.

Proposed Solution:

- Use **Capacity Scheduler** to divide resources among ETL, reporting, and ML queues.
- Give the **ETL queue higher priority** because it is time-sensitive.
- Reserve minimum resources for reporting and ML to maintain fairness.
- Enable YARN's automatic task/container restart when a node fails.
- Configure ResourceManager High Availability for better recovery.

Example Policy:

Queue	Priority	Resource Allocation
ETL	High	50%
Reporting	Medium	25%
Machine Learning	Medium	25%

Justification:

This policy ensures ETL jobs receive priority while other workloads still receive guaranteed resources. Automatic recovery improves cluster reliability.

4. Integrating SAN, NAS and Operational Databases

Problem:

An enterprise needs to ingest data from **SAN, NAS, and operational databases** while ensuring reliable backup and avoiding unnecessary duplication.

Proposed Architecture:

SAN / NAS / Databases

↓

Apache NiFi / ETL Layer

↓

Data Validation & Deduplication

↓

Cloud Data Lake / HDFS

↓

Backup & Disaster Recovery

Components:

- **Apache NiFi:** Collects and routes data from different sources.
- **ETL tools:** Extract, transform, and validate operational data.
- **Deduplication:** Identifies repeated records before storage.
- **Cloud Data Lake/HDFS:** Provides scalable centralized storage.
- **Backup storage:** Maintains copies for disaster recovery.

Justification:

The architecture provides centralized ingestion, reliable backups, and controlled duplication while allowing different enterprise data sources to be integrated efficiently.

5. End-to-End Hadoop Ecosystem Solution

Problem:

A big data platform must provide **secure data movement, high availability, and Apache NiFi/ETL integration**.

Proposed Architecture:

Data Sources

↓

Apache NiFi

↓

Secure Transfer (TLS/Encryption)

↓

HDFS / Cloud Data Lake

↓

YARN + Spark/MapReduce

↓

Hive / Analytics



Applications & Reports

Key Components:

- **Apache NiFi:** Ingests, routes, and monitors data flows.
- **TLS/Encryption:** Protects data during transmission.
- **HDFS:** Provides distributed and fault-tolerant storage.
- **YARN:** Manages cluster resources and applications.
- **Spark/MapReduce:** Performs distributed processing.
- **Hive:** Supports SQL-based big data analysis.
- **NameNode HA and data replication:** Provide high availability.

Justification:

The architecture securely moves data from multiple sources, provides fault-tolerant storage and processing, and integrates NiFi with the Hadoop ecosystem. Therefore, it satisfies all three constraints: **security, high availability, and ETL integration.**

Constraint-Based Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Constraints	25%	All technical constraints are identified and prioritized accurately	Most constraints identified	Some constraints identified	Constraints poorly understood
System Design Quality	25%	Design is robust, feasible, and aligned to constraints	Reasonable design	Basic design with gaps	Weak design
Application of Hadoop Concepts	20%	Uses HDFS, MapReduce, YARN, and integration concepts effectively	Mostly effective use	Partial use of concepts	Weak concept application
Justification	15%	Strong technical reasoning for all choices	Good justification	Basic justification	Little justification
Clarity	15%	Clear, well-structured, and professional answer	Mostly clear	Somewhat unclear	Poorly structured